AI Express Hackathon — Technical Summary Sheet

Course Code	BCA318-5 (AI Express Hackathon)	**Group ID**	GROUP 8
Members	Member 1- Simran Rao Member 2- Zehrah Shameer Member 3 - Sasha Alwin	**Track**	Track 3 - Legal Compliance Drone (Unit 4, First-Order Logic Agent)
GitHub Repository	https://github.com/zehrahshameer07/Legal-Compliance-Drone		

1. PEAS Framework

Performance Measure	Zero illegal airspace incursions; shortest legal path length; total mission time
Environment	10x7 partially-observable urban grid; static (but initially hidden) airspace-restriction map
Actuators	Move N/E/S/W; sensor sweep; execute FOL backward-chaining query
Sensors	Local zone inspector — reveals restriction/permit status of the next cell only

2. Core Algorithmic Formulation

Constants: Drone; Z_x_y per grid cell. **Variable:** ?z (universally quantified). **Ground facts** asserted only once a cell is sensed.

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R1: Restricted(z) ∧ ¬HasPermit(Drone,z) ⇒ ¬FlyOver(Drone,z)
R2: ¬Restricted(z) ⇒ FlyOver(Drone,z)
R3: Restricted(z) ∧ HasPermit(Drone,z) ⇒ FlyOver(Drone,z)
```

State space: agent position (x,y) on the grid. **Initial state:** start pad (0,3). **Goal test:** position == goal pad (9,3). **Path cost:** 1 per legal move (BFS, uniform cost). **Decision rule per cell:** before every move, backward-chain the query FlyOver(Drone,z)? against the rule base above; GRANTED → move, DENIED → mark blocked and re-run BFS from the current cell (dynamic replanning). Forward chaining materialises FlyOver/¬FlyOver facts immediately after each new sensor reading.

3. Complexity Analysis

Component	Theoretical	Observed (this run)
Unification	O(n), n = term size	n ≤ 3 (fixed-arity predicates)
Forward chaining (per pass)	O(rules x facts^premises)	≤ 3 rules, small fact base → near-linear
Backward chaining (FOL-BC-ASK)	O(branching^rule-depth)	rule depth ≤ 2 → O(1) per query
Path (re)planning	O(V+E) = O(W x H) per BFS	45-60 nodes expanded per plan
Space	O(cells sensed + path length)	≤ 70 cells, path ≤ 15 cells

Sample end-to-end run: 15 moves, 18 FOL queries executed, 3 legal denials, 4 replans, 220 total BFS nodes expanded, < 0.05s compute time.

Deliverable behaviour: the drone pauses before every new zone, prints a full backward-chaining proof trace, and only crosses into a zone once FlyOver(Drone,z) is proven GRANTED — verified in main_pygame.py / main_ascii.py console + on-screen log.